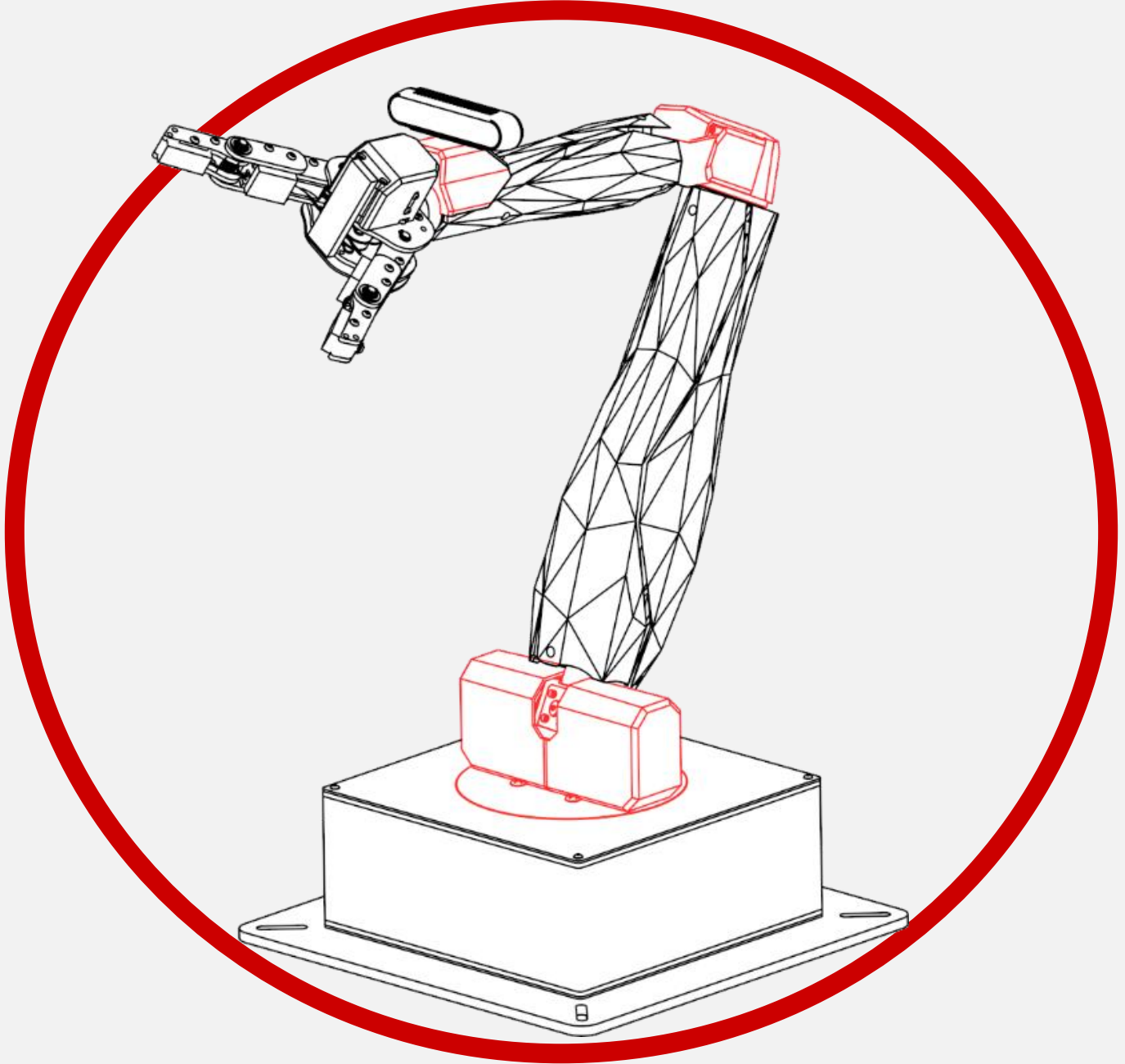




QArm

Object Detection

V1.0 – 13th February 2025

© 2025 Quanser Consulting Inc., All rights reserved.
For more information on the solutions Quanser offers,
please visit the web site at: <http://www.quanser.com>



Quanser Consulting Inc. info@quanser.com
119 Spy Court Phone : 19059403575
Markham, Ontario Fax : 19059403576
L3R 5H6, Canada printed in Markham, Ontario.

This document and the software described in it are provided subject to a license agreement. Neither the software nor this document may be used or copied except as specified under the terms of that license agreement. Quanser Consulting Inc. ("Quanser") grants the following rights: a) The right to reproduce the work, to incorporate the work into one or more collections, and to reproduce the work as incorporated in the collections, b) to create and reproduce adaptations provided reasonable steps are taken to clearly identify the changes that were made to the original work, c) to distribute and publicly perform the work including as incorporated in collections, and d) to distribute and publicly perform adaptations. The above rights may be exercised in all media and formats whether now known or hereafter devised. These rights are granted subject to and limited by the following restrictions: a) You may not exercise any of the rights granted to You in above in any manner that is primarily intended for or directed toward commercial advantage or private monetary compensation, and b) You must keep intact all copyright notices for the Work and provide the name Quanser for attribution. These restrictions may not be waived without express prior written permission of Quanser.



Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

QArm – Application Guide

Object Detection

Why explore Object Detection?

After understanding different color spaces and how images are processed digitally, the next step is to use these concepts for real life applications. This lab will focus on object detection, which allows us to identify and locate elements of interest from an image or a video. In this lab, you will learn the basics of different image filters as well as how to start detecting objects in the image. Future labs will focus on how to use this information to manipulate the QArm to find or follow selected objects in its workspace.

Background

The QArm content contains 3 labs that focus on visual servoing. The first one focuses on learning how to do image acquisition, the second one in object detection, and the last one focuses on visual servoing, which is moving the arm based on what the camera sees. This lab focuses on identifying our object of choice.

Prior to starting this lab, please review the following concept reviews (should be located in Documents/Quanser/4_concept_reviews/),

- Concept Review – Image Filters (Everything before the Sharpening Section).
- Concept Review – Blob Detection

Getting started

The goal of this lab is to use the QArm camera to find the blob of the object of interest. A Simulink model will be used to read the image, add filters and finally find the blob.

Ensure you have completed the following labs

- **Image Acquisition Lab**

Before you begin this lab, ensure that the following criteria are met.

- The QArm has been setup and tested. See the QArm Quick Start Guide for details on this step.
- You are familiar with the basics of Simulink. See the [Simulink Onramp](#) for more help with getting started with Simulink.